SMART SENSOR TECHNOLOGIES		Semester	7
Course Code	**BCO714A**	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		

Course objectives:

- Understand the fundamentals of smart sensors and sensing technologies
- Familiarize with various application specific sensors
- Learn sensors with microcontrollers and wireless sensing
- Study the real-world applications of smart sensors

Teaching-Learning Process (General Instructions)

These are sample strategies; which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes.
2. Utilize video/animation films to illustrate the functioning of various concepts.
3. Promote collaborative learning (Group Learning) in the class.
4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking.
5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it.
6. Introduce topics through multiple representations.
7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions.
8. Discuss the real-world applications of every concept to enhance students' comprehension.
9. Use any of these methods: Chalk and board, Active Learning, Case Studies.

Module-1

Smart Sensor Basics: Introduction, Sensor Vs Transducer, Nature of Sensors, Sensor Output Characteristics, Sensing Technologies, Digital Output Sensors.

Textbook 1: Ch. 1.1, 1.3, 3.2-3.4
Textbook 2: Ch. 1.1

Module-2

Application Specific Sensors: Ultrasonic Detectors, Microwave Motion Detectors, Capacitive Occupancy Detectors, Optical Presence Sensors, Light Detectors: Photo Diode, Phototransistor, Photoresistor, CCD and CMOS Imaging Sensors, Temperature Sensors: Thermoelectric Sensors, Optical Temperature Sensors.

Textbook 2: Ch. 7.1, 7.2, 7.6, 7.9, 15.2-15.4, 15.6, 17.8, 17.9

Module-3

Sensor with Microcontroller: Introduction, Amplification and Signal Conditioning, Integrated Signal Conditioning, Digital Conversion, MCU Control, MCUs for Sensor Interface, Techniques and Systems Considerations, Sensor Integration.

Textbook 1: Ch. 4.1, 4.2, 4.3.1, 4.4, 5.2, 5.3, 5.5, 5.7

Module-4

Wireless Sensing: Wireless Data and Communications, Wireless Sensing Networks, Industrial Wireless Sensing Networks, RF Sensing, Telemetry, RF MEMS, Application Example.

Textbook 1: Ch. 8.2-8.8

Module-5

Smart Applications and System Requirements: Automotive Applications, Industrial (Robotic) Applications, Consumer Applications, Future Sensor Plus Semiconductor Capabilities, Future System Requirements, Software-Sensing and the System, Trusted Sensing, Alternate Views of Smart Sensing.

Textbook 1: Ch 14.3-14.5, 16.2-16.6

Course Outcomes (Course Skill Set)

At the end of the course, the student will be able to:

1. Explain the basics of smart sensors and various sensing technologies.
2. Illustrate the role of various application-specific sensors and occupancy detectors.
3. Demonstrate the interfacing of different types of sensors with Microcontrollers.
4. Infer Wireless Sensing, RF Sensing and RF MEMS with examples.
5. Identify the various real-time applications of smart sensors.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.

2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Book:

1. Randy Frank, "Understanding Smart Sensors", Artech House Integrated Microsystems Series, 3rd Edition.
2. Jacob Fraden, "Handbook of Modern Sensors: Physics, Designs, and Applications", 5th Edition, Springer, 2016.

Reference Books:

1. Vlasios Tsiatsis, Stamatis Karnouskos, Jan Holler, David Boyle, Catherine Mulligan, "Internet of Things: Technologies and Applications for a New Age of Intelligence", Academic Press, 2018.
2. Henry Leung, Subhas Chandra Mukhopadhyay, "Intelligent Environmental Sensing", Springer, 2015.

Web links and Video Lectures (e-Resources):

- https://www.youtube.com/watch?v=5b5xJu8KYrc
- https://www.youtube.com/watch?v=oRydUfgMdgA
- https://www.youtube.com/watch?v=i0_gJFYyb2g
- https://nptel.ac.in/courses/108106193

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Course project (Group ot two) to develop real-time applications of smart sensors in various fields like home automation, healthcare, environmental monitoring, agriculture, smart cities etc. (25 marks)

IOT PLATFORMS AND SYSTEM DESIGN		Semester	7
Course Code	**BCO714B**	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		

Course objectives:

- Understand physical & logical design of IoT and system management using NETCONF-YANG
- Learn practical skills to design IoT systems using Python for logical design and interface with physical IoT devices like Raspberry Pi
- Study different cloud storage models and communication APIs relevant to IoT and web frameworks for building IoT applications
- Study the real-world applications and case studies of IoT across diverse domains
- Introduce big data analytics tools and frameworks, such as Hadoop and Spark, in the context of processing and analyzing data generated by IoT devices

Teaching-Learning Process (General Instructions)

These are sample strategies; which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes.
2. Utilize video/animation films to illustrate the functioning of various concepts.
3. Promote collaborative learning (Group Learning) in the class.
4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking.
5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it.
6. Introduce topics through multiple representations.
7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions.
8. Discuss the real-world applications of every concept to enhance students' comprehension.
9. Use any of these methods: Chalk and board, Active Learning, Case Studies.

Module-1

Introduction and Concepts: Physical Design of IoT, Logical design of IoT, IoT enabling Technologies, IoT levels and Deployment Templates.

IoT System Management with NETCONF-YANG: Need for IoT Systems Management, Simple Network Management Protocol (SNMP), Network Operator Requirements, NETCONF, YANG, IoT Systems Management with NETCONF-YANG.

Textbook: Ch. 1.2-1.5, Ch. 4.1-4.6

Module-2

IoT Platforms Design Methodology: Introduction, IoT Design Methodology, Case Study on IoT System for Weather Monitoring, Motivation for Using Python.

IoT Systems-Logical Design using Python: Python Packages of Interest for IoT.

IoT Physical Devices & Endpoints: Basics of IoT Device, Exemplary Device: Raspberry PI, About the Board, Linux on Raspberry PI, Raspberry PI Interfaces, Programming Raspberry PI with Python, Other IoT Devices.

Textbook: Ch. 5.1-5.4, Ch. 6.11, Ch. 7.1-7.7

Module-3

IoT Physical Servers & Cloud Offerings: Cloud Storage Models & Communication APIs, WAMP-AutoBahn for IoT, Xively Cloud for IoT, Python Web Application Framework-Django, Designing a RESTful Web API, Amazon Web Services for IoT, SkyNet IoT Messaging Platform.

Textbook: Ch. 8.1-8.7

Module-4

Case Studies Illustrating IoT Design: Introduction, Home Automation, Cities, Environment, Agriculture, Productivity Applications.

Textbook: Ch. 9.1-9.6

Module-5

Data Analytics for IoT: Introduction, Apache Hadoop, Using Hadoop MapReduce for Batch Data Analysis, Apache Oozie, Apache Spark, Apache Storm, Using Apache Storm for Real-time Data Analysis, Structural Health Monitoring Case Study.

Textbook 1: Ch 10.1-10.8

Course Outcomes (Course Skill Set)

At the end of the course, the student will be able to:

1. Explain the core design components of IoT systems and the role of NETCONF-YANG in managing IoT devices and networks.
2. Apply IoT design methodologies to design IoT systems using Python and Raspberry Pi to interact with sensors and actuators.
3. Choose appropriate cloud services and web frameworks to build IoT applications.
4. Build IoT solutions for various real-world scenarios, drawing insights from diverse case studies in areas like home automation and smart cities etc.
5. Utilize big data tools like Hadoop and Storm to perform batch and real-time data analysis on IoT-generated data, understanding their respective applications and benefits.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered

- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Book:

1. Arshdeep Bahga, Vijay Madisetti "Internet of Things: A Hands-on Approach", Universities Press (India) Publications.

Reference Books:

1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry,"IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things", 1st Edition, Pearson Education (Cisco Press Indian Reprint). (ISBN: 978-9386873743).
2. Raj Kamal, "Internet of Things", Second Edition, McGraw Hill India, 2022.

Web links and Video Lectures (e-Resources):

- https://www.youtube.com/watch?v=bdCdKdfDf7o
- https://www.youtube.com/watch?v=cSIcqxvgChQ
- https://www.youtube.com/watch?v=xMkWCvlCC4Y
- https://onlinecourses.nptel.ac.in/noc22_cs53/preview

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Course project (group of two): design and develop real-time IoT applications in various fields like home automation, healthcare, environmental monitoring, agriculture, smart cities, productive applications etc. (25 marks)

IOT AUTOMATION		Semester	7
Course Code	**BCO714C**	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		

Course objectives:

- Understand the evolution of industrial automation from traditional systems to modern IoT and System of Systems (SoS) architectures
- Grasp the foundational concepts of Local Automation Clouds, including their properties, establishment and application engineering
- Study engineering methodologies for designing and implementing IoT automation systems
- Learn application automation systems for complex industrial and societal challenges

Teaching-Learning Process (General Instructions)

These are sample strategies; which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes.
2. Utilize video/animation films to illustrate the functioning of various concepts.
3. Promote collaborative learning (Group Learning) in the class.
4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking.
5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it.
6. Introduce topics through multiple representations.
7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions.
8. Discuss the real-world applications of every concept to enhance students' comprehension.
9. Use any of these methods: Chalk and board, Active Learning, Case Studies.

Module-1

Towards Industrial and Societal Automation and Digitization: The Need of New Technology, From DCS and SCADA to IoT and System of Systems, Automation System Architectures, Current Trends in Automation Systems, Future Automation System Requirements, Next Generation Automation and Digitization Technology-IoT and SoS, The Local Automation Cloud Approach.

Local Automation Clouds: The Local Cloud Concept, Local Cloud Properties, Local Cloud Establishment, Automation Support, Automation Application Engineering in Local Clouds, Latency in Local Clouds, Security in Local Clouds, SoS Scalability.

Textbook: Ch. 1.1-1.7, Ch. 2.1-2.8

Module-2

Engineering of IoT Automation Systems: Introduction, Component-based Engineering Methodology, Safety and Security Engineering of IoT Automation Systems, Engineering

Scenarios: Efficient Deployment of a Large Number of IoT Sensors, Swift Deployment and Configuration.

Textbook: Ch. 6.1, 6.3, 6.4, 6.5.1, 6.5.2

Module-3

Application System Design–Energy Optimization: Introduction, Market as an Energy Optimizing Method, Optimizations Based on a Virtual Market of Energy, Energy Optimization on Lifts, Context Aware Streets, Optimization of Municipal Service Systems.

Textbook: Ch. 7.1-7.7

Module-4

Application System Design-Complex Systems Management and Automation: Introduction, Vision from an Automation Perspective, Electromobility Systems of Systems, Electric Vehicles and Recharge Infrastructure, Arrowhead SOA EM Solution, Systems and Services, Arrowhead EM Services and Related Automation Aspects, Co-simulation Platform.

Textbook: Ch. 9.1-9.9

Module-5

Application System Design-High Security: Introduction, Smart Maintenance Use Case, Authentication and Certification Service, ISO/IEC 20922 Support for the Smart Service Architecture, Safety and Security Analysis for Identifying System Vulnerabilities.

Application System Design-Smart Production: Automotive Manufacturing, Manufacturing of Electrical Cabinets, Asset localization in Mines.

Textbook 1: Ch 10.1-10.5, Ch 11.1-11.3

Course Outcomes (Course Skill Set)

At the end of the course, the student will be able to:

1. Compare traditional automation architectures (DCS, SCADA) and modern approaches like IoT and System of Systems (SoS).
2. Explain the architecture and functionalities of Local Automation Clouds to evaluate their applicability and limitations in various automation contexts,
3. Develop strategies for the safe and secure engineering of IoT automation systems, including efficient deployment and configuration of sensors and devices.
4. Identify solutions for energy optimization in industrial and urban environments by applying concepts like virtual energy markets and context-aware systems.
5. Outline case studies related to complex systems management and high-security automation etc., demonstrating the ability to integrate advanced automation concepts for real-world problems.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment

Test component, there are 25 marks.

- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Book:

1. Jerker Delsing, "IoT Automation: Arrowhead Framework", CRC Press, Taylor & Francis Group.

Reference Books:

1. The Internet of Things in the Industrial Sector, Mahmood, Zaigham (Ed.) (Springer Publication).
2. Industrial Internet of Things: Cyber manufacturing System, Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat (Springer Publication).

Web links and Video Lectures (e-Resources):

- https://www.youtube.com/watch?v=HmbUJEShA-8
- https://www.youtube.com/watch?v=eidD14dXW8s
- https://www.udemy.com/course/iot-internet-of-things-automation-with-esp8266/?srsltid=AfmBOoqqKDLk_fKye1CPIO9QjctFfgHOf1F_Zkmtze8uNRUvmq_9s1-3
- https://onlinecourses.nptel.ac.in/noc20_cs69/preview

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Course project (Group of two): Develop real-time applications of IoT Automation in various fields like home automation, healthcare, environmental monitoring, agriculture, smart cities etc. (25 marks)

Deep Learning			Semester	7
Course Code	**BCS714A**		CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0		SEE Marks	50
Total Hours of Pedagogy	40		Total Marks	100
Credits	03		Exam Hours	03
Examination type (SEE)	Theory			

Course objectives:

- Understand the basic concepts of deep learning.
- Know the basic working model of Convolutional Neural Networks and RNN in decision making.
- Illustrate the strength and weaknesses of many popular deep learning approaches.
- Introduce major deep learning algorithms, the problem settings, and their applications to solve real world problems

Teaching-Learning Process (General Instructions)

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation/Demonstration to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem/Practical Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills, and practical skill such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
6. Use animations/videos to help the students to understand the concepts.
7. Demonstrate the concepts using PYTHON and its libraries wherever possible

Module-1

Introducing Deep Learning: **Biological and Machine Vision:** Biological Vision, Machine Vision: The Neocognitron, LeNet-5, The Traditional Machine Learning Approach, ImageNet and the ILSVRC, AlexNet, TensorFlow Playground. **Human and Machine Language:** Deep Learning for Natural Language Processing: Deep Learning Networks Learn Representations Automatically, Natural Language Processing, A Brief History of Deep Learning for NLP, Computational Representations of Language: One-Hot Representations of Words, Word Vectors, Word-Vector Arithmetic, word2viz, Localist Versus Distributed Representations, Elements of Natural Human Language.

Text book 2 : Chapter 1, 2

Module-2

Regularization for Deep Learning: Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi- Supervised Learning, Multi-Task Learning, Early Stopping, Parameter Tying and Parameter Sharing, Sparse Representations, **Optimization for Training Deep Models:** How Learning Differs from Pure Optimization, Basic Algorithms. Parameter Initialization Strategies, Algorithms with Adaptive Learning Rates.

Text book 1 : Chapter 7 (7.1 to 7.10), Chapter 8 (8.1,8.3,8.4,8.5)

Module-3

	Convolution neural networks: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Convolutional Networks and the History of Deep Learning. **Text book 1 : Chapter 9 (9.1 to 9.8, 9.11)**
	Module-4
	Sequence Modelling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks. Long short-term memory. **Text book 1 : Chapter 10 (10.1 to 10.6, 10.10)**
	Module-5
	Interactive Applications of Deep Learning: Natural Language Processing: Preprocessing Natural Language Data: Tokenization, Converting All Characters to Lowercase, Removing Stop Words and Punctuation, Stemming, Handling *n*-grams, Preprocessing the Full Corpus, Creating Word Embeddings with word2vec: The Essential Theory Behind word2vec, Evaluating Word Vectors, Running word2vec, Plotting Word Vectors, The Area under the ROC Curve: The Confusion Matrix, Calculating the ROC AUC Metric, Natural Language Classification with Familiar Networks: Loading the IMDb Film Reviews, Examining the IMDb Data, Standardizing the Length of the Reviews, Dense Network, Convolutional Networks, Networks Designed for Sequential Data: Recurrent Neural Networks, Long Short-Term Memory Units, Bidirectional LSTMs, Stacked Recurrent Models, Seq2seq and Attention, Transfer Learning in NLP. **Text book 2 : Chapter-8**

Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1. Interpret the concepts of neural networks learning processes.
2. Illustrate deep learning methods using regularization and Optimization process
3. Design deep learning models using convolutional operations.
4. Analyze sequential data to build recurrent and recursive models.
5. Demonstrate the different interactive applications of deep learning.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Books

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 2016. https://www.deeplearningbook.org/lecture_slides.html

2. John Krohn, Grant Beyleveld, Aglae Bassens, Deep Learning Illustrated, A Visual, Interactive Guide to Artificial Intelligence, Pearson, 2022.

Web links and Video Lectures (e-Resources):

https://www.youtube.com/watch?v=VyWAvY2CF9c
https://www.youtube.com/watch?v=7sB052Pz0sQ
https://www.youtube.com/watch?v=Mubj_fqiAv8
https://www.coursera.org/learn/neural-networks-deep-learning

- **https://onlinecourses.nptel.ac.in/noc20_cs62/preview**

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Programming Assignments, such as implementation of CNN and Recurrent neural network models - 10 Marks
- Group assignment (Group of two) on recent developments in Deep learning – Refer IEEE/ACM/Elsevier etc publications - 15 Marks

Introduction to DBMS		Semester	7
Course Code	**BCS755A**	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	3
Examination nature (SEE)	Theory		

Course objectives:

- To Provide a strong foundation in database concepts, technology, and practice.
- To Practice SQL programming through a variety of database problems.
- To Understand the relational database design principles.
- To Demonstrate the use of concurrency in database.
- To Design and build database applications for real world problems.

Teaching-Learning Process (General Instructions)

These are sample Strategies; that teachers can use to accelerate the attainment of the various course outcomes.

- Lecturer method (L) needs not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
- Use of Video/Animation to explain functioning of various concepts.
- Encourage collaborative (Group Learning) Learning in the class.
- Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
- Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.
- Use any of these methods: Chalk and board, Active Learning, Case Studies.

MODULE-1

Introduction to Databases: Introduction, Characteristics of database approach, Advantages of using the DBMS approach, History of database applications.

Overview of Database Languages and Architectures: Data Models, Schemas, and Instances. Three schema architecture and data independence, database languages, and interfaces, The Database System environment.

Textbook 1:Ch 1.1 to 1.8, 2.1 to 2.6

MODULE-2

Conceptual Data Modeling using Entities and Relationships: Entity types, Entity sets and structural constraints, Weak entity types, ER diagrams, Specialization and Generalization.

Mapping Conceptual Design into a Logical Design: Relational Database Design using ER-to-Relational mapping

Textbook 1: Ch 3.1 to 3.10, 9.1 & 9.2

MODULE-3

Relational Model: Relational Model Concepts, Relational Model Constraints and relational database schemas, Update operations, transactions, and dealing with constraint violations.

Relational Algebra: Unary and Binary relational operations, additional relational operations (aggregate, grouping, etc.) Examples of Queries in relational algebra.

Textbook 1: Ch 5.1 to 5.3, Ch 8.1 to 8.5

MODULE-4

SQL: SQL data definition and data types, Schema change statements in SQL, specifying constraints in SQL, retrieval queries in SQL, INSERT, DELETE, and UPDATE statements in SQL, Additional features of SQL

Normalization: Database Design Theory – Introduction to Normalization using Functional and Multivalued Dependencies: Informal design guidelines for relation schema, Functional Dependencies, Normal Forms based on Primary Keys, Second and Third Normal Forms, Boyce-Codd Normal Form, Multivalued Dependency and Fourth Normal Form, Join Dependencies and Fifth Normal Form.

Textbook 1: Ch 6.1 to 6.5,14.1 to 14.7

MODULE-5

SQL: Advanced Queries: More complex SQL retrieval queries, Specifying constraints as assertions and action triggers, Views in SQL.

Concurrency Control in Databases: Two-phase locking techniques for Concurrency control, Concurrency control based on Timestamp ordering, Multiversion Concurrency control techniques, Validation Concurrency control techniques, Granularity of Data items and Multiple Granularity Locking.

Textbook 1: Ch 7.1 to 7.3, 21.1 to 21.5

Course outcomes (Course Skill Set):
At the end of the course, the student will be able to:

- Demonstrate the basic elements of a database management system.
- Design entity relationship and convert entity relationship diagrams into RDBMS.
- Use Structured Query Language (SQL) for database manipulation.
- Apply normalization to increase the efficiency of database design.
- Illustrate the concepts of concurrency control techniques.

Assessment Details (both CIE and SEE)
The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:
For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.
Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

- Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours).
- The question paper will have ten questions. Each question is set for 20 marks.
- There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module.
- The students have to answer 5 full questions, selecting one full question from each module.
- Marks scored shall be proportionally reduced to 50 marks.

Suggested Learning Resources:

Text Books:

1. Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 7th Edition, 2017, Pearson.

Reference Books:

1. Database management systems, Ramakrishnan, and Gehrke, 3rd Edition, 2014, McGraw Hill

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

Course Project (25 marks)

- For any problem selected
    - Develop the application having at least five tables & domain areas shall include health care, agriculture & so on.

	Introduction to Algorithms		Semester	7
	Course Code	BCS755B	CIE Marks	50
	Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
	Total Hours of Pedagogy	40	Total Marks	100
	Credits	03	Exam Hours	03
	Examination type (SEE)	Theory		

Course objectives:

- To learn the methods for analyzing algorithms and evaluating their performance.
- To demonstrate the efficiency of algorithms using asymptotic notations.
- To solve problems using various algorithm design methods, including brute force, greedy, divide and conquer, decrease and conquer, transform and conquer, dynamic programming, backtracking, and branch and bound.
- To learn the concepts of P and NP complexity classes.

Teaching-Learning Process (General Instructions)

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes.
2. Utilize video/animation films to illustrate the functioning of various concepts.
3. Promote collaborative learning (Group Learning) in the class.
4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking.
5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it.
6. Introduce topics through multiple representations.
7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions.
8. Discuss the real-world applications of every concept to enhance students' comprehension.

Module-1

INTRODUCTION: What is an Algorithm?, Fundamentals of Algorithmic Problem Solving, Important problem Types, Fundamental Data Structures, Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, ,Analysis Framework, Asymptotic Notations and Basic Efficiency Classes,

Chapter 1 (Sections 1.1 to 1.4), Chapter 2 (2.1, 2.2)

Module-2

FUNDAMENTALS OF THE ANALYSIS OF ALGORITHM EFFICIENCY: Mathematical Analysis of Non-recursive Algorithms, Mathematical Analysis of Recursive Algorithms.

BRUTE FORCE APPROACHES: Selection Sort and Bubble Sort, Sequential Search and Brute Force String Matching.

Chapter 2(Sections 2.3,2.4), Chapter 3(Section 3.1,3.2)

	Module-3
	Exhaustive Search (Travelling Salesman problem and Knapsack Problem). **Depth First search and Breadth First search.** **DECREASE-AND-CONQUER:** Insertion Sort, Topological Sorting. **DIVIDE AND CONQUER:** Merge Sort, Binary Tree Traversals. **Chapter 3(3.4,3.5), Chapter 4 (Sections 4.1,4.2), Chapter 5 (Section 5.1,5.3)**
	Module-4
	TRANSFORM-AND-CONQUER: Balanced Search Trees (AVL Trees), Heaps and Heapsort. **SPACE-TIME TRADEOFFS:** Sorting by Counting: Comparison counting sort, Input Enhancement in String Matching: Horspool's Algorithm, Hashing. **Chapter 6 (Sections 6.3,6.4), Chapter 7 (Sections 7.1,7.2, 7.3)**
	Module-5
	DYNAMIC PROGRAMMING: Three basic examples, The Knapsack Problem and Memory Functions. **THE GREEDY METHOD:** Kruskal's Algorithm, Dijkstra's Algorithm, Huffman Trees and Codes. **Chapter 8 (Sections 8.1,8.2), Chapter 9 (Sections 9.2,9.3,9.4)**

Course outcome (Course Skill Set)

At the end of the course, the student will be able to:

1. Explain the algorithm design steps and computational problem types.
2. Apply the asymptotic notational method to analyze the performance of the algorithms in terms of time complexity.
3. Demonstrate divide & conquer approaches and decrease & conquer approaches to solve computational problems.
4. Make use of the transform & conquer design approach to solve the given real-world or complex computational problems.
5. Apply greedy and dynamic programming methods to solve graph & string-based computational problems.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
- The first test will be administered after 40-50% of the syllabus has been covered, and the second test will be administered after 85-90% of the syllabus has been covered
- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Textbooks

1. Introduction to the Design and Analysis of Algorithms, By Anany Levitin, 3rd Edition (Indian), 2017, Pearson.

Reference books

1. Computer Algorithms/C++, Ellis Horowitz, SatrajSahni and Rajasekaran, 2nd Edition, 2014, Universities Press.
2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI.
3. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)

Web links and Video Lectures (e-Resources):

- Design and Analysis of Algorithms: https://nptel.ac.in/courses/106/101/106101060/

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

1. Problem Solving - Competitive programming (Hacker Rank/ Hacker Earth / Leetcode) – 10 Marks
2. Problem solving (Numerical examples) related to different algorithms – 15 Marks

	SOFTWARE ENGINEERING		Semester	7
	Course Code	**BCS755C**	CIE Marks	50
	Teaching Hours/Week (L:T:P: S)	3:0:0:0	SEE Marks	50
	Total Hours of Pedagogy	50	Total Marks	100
	Credits	04	Exam Hours	3
	Examination type (SEE)	Theory		

Course objectives:

To understand foundational principles and the evolving nature of software engineering.
- To learn various software process models and their practical applications.
- To acquire skills in gathering, modeling, and validating software requirements.
- To apply Agile methodologies and understand core software engineering practices.
- To build a foundation for software design, testing, and quality assurance.

Teaching-Learning Process

These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) needs not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.
6. Introduce Topics in manifold representations.
7. Show the different ways to solve the same problem with different circuits/logic and encourage the students to come up with their own creative ways to solve them.
8. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding
9. Use any of these methods: Chalk and board, Active Learning, Case Studies

Module-1

Software and Software Engineering: The nature of Software, The unique nature of WebApps, Software Engineering, The software Process, Software Engineering Practice, Software Myths.

Process Models: A generic process model, Process assessment and improvement, Prescriptive process models: Waterfall model, Incremental process models, Evolutionary process models, Concurrent models, Specialized process models. Unified Process , Personal and Team process models

Textbook 1: Chapter 1: 1.1 to 1.6, Chapter 2: 2.1 to 2.5

Module-2

Understanding Requirements: Requirements Engineering, Establishing the ground work, Eliciting Requirements, Developing use cases, Building the requirements model, Negotiating Requirements, Validating Requirements.

Requirements Modeling Scenarios, Information and Analysis classes: Requirement Analysis, Scenario based modeling, UML models that supplement the Use Case, Data modeling Concepts, Class-Based Modeling.

Requirement Modeling Strategies : Flow oriented Modeling , Behavioral Modeling.

Textbook 1: Chapter 5: 5.1 to 5.7, Chapter 6: 6.1 to 6.5, Chapter 7: 7.1 to 7.3

Module-3

Agile Development: What is Agility?, Agility and the cost of change. What is an agile Process?, Extreme Programming (XP), Other Agile Process Models, A tool set for Agile process .

Principles that guide practice: Software Engineering Knowledge, Core principles, Principles that guide each framework activity.

Textbook 1: Chapter 3: 3.1 to 3.6, Chapter 4: 4.1 to 4.3

Module-4

Software Design: Design within the context of software engineering, Design process and quality, Design concepts: abstraction, modularity, architecture, patterns.

Architectural Design: Architectural styles and patterns, reference architectures, component-level design, designing class-based components, conducting component-level design, design for reuse.

Textbook 1:Chapter 8: 8.1–8.6, Chapter 9: 9.1–9.5

Module-5

Software Testing: Introduction to software testing, Strategic approach, Test strategies for conventional and object-oriented software, Validation testing, System testing, White-box and Black-box testing, Basis Path Testing, Control structure testing.

Software Quality: Concepts of quality, Software quality assurance, Reviews, Software reliability and metrics.

Textbook 1: Chapter 14: Sections 14.1 to 14.5,Chapter 15: Sections 15.1 to 15.5, Chapter 19: Sections 19.1 to 19.5

Course outcome

At the end of the course, the student will be able to :

1. Explain the software nature, engineering practices, myths, and software process models.
2. Apply requirements engineering, elicitation, modeling, and validation in software development.
3. Demonstrate agile principles, practices, and tools for software development agility.
4. Apply design concepts, process, and architecture for quality software development.
5. Explain software testing strategies and quality assurance for reliable software.

Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

Continuous Internal Evaluation:

- For the Assignment component of the CIE, there are 25 marks and for the Internal Assessment Test component, there are 25 marks.
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- Any two assignment methods mentioned in the 22OB2.4, if an assignment is project-based then only one assignment for the course shall be planned. The teacher should not conduct two assignments at the end of the semester if two assignments are planned.
- For the course, CIE marks will be based on a scaled-down sum of two tests and other methods of assessment.

Internal Assessment Test question paper is designed to attain the different levels of Bloom's taxonomy as per the outcome defined for the course.

Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

1. The question paper will have ten questions. Each question is set for 20 marks.
2. There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), **should have a mix of topics** under that module.
3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Textbook

Roger S. Pressman: Software Engineering – A Practitioner's Approach, 7th Edition, Tata McGraw Hill, 2010.

Web links and Video Lectures (e-Resources):

https://www.geeksforgeeks.org/software-engineering/software-engineering/

Activity-Based Learning (Suggested Activities in Class)/Practical-Based learning

- Course project (Group of two students): Simulation that covers all the phases of SDLC - 25 marks

INTRODUCTION TO EMBEDDED SYSTEMS		Semester	7
Course Code	**BCE755D**	CIE Marks	50
Teaching Hours/Week (L: T:P: S)	3:0:0:0	SEE Marks	50
Total Hours of Pedagogy	40	Total Marks	100
Credits	03	Exam Hours	03
Examination type (SEE)	Theory		

Course objectives:

- Identify the components, purpose and applications of the Embedded Systems
- Learn the RTOS and IDE for Embedded System Design
- Understand the fundamentals of ARM-based systems and basic architecture of CISC and RISC
- Familiarize with ARM programming modules along with registers, CPSR and Flags

Teaching-Learning Process (General Instructions)

These are sample strategies; which teachers can use to accelerate the attainment of the various course outcomes.

1. Lecturer method (L) does not mean only the traditional lecture method, but different types of teaching methods may be adopted to achieve the outcomes.
2. Utilize video/animation films to illustrate the functioning of various concepts.
3. Promote collaborative learning (Group Learning) in the class.
4. Pose at least three HOT (Higher Order Thinking) questions in the class to stimulate critical thinking.
5. Incorporate Problem-Based Learning (PBL) to foster students' analytical skills and develop their ability to evaluate, generalize, and analyze information rather than merely recalling it.
6. Introduce topics through multiple representations.
7. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions.
8. Discuss the real-world applications of every concept to enhance students' comprehension.
9. Use any of these methods: Chalk and board, Active Learning, Case Studies.

Module-1

Introduction to Embedded Systems: What is an Embedded System? Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification of Embedded systems, Major Application Areas of Embedded Systems. Purpose of Embedded Systems.

The Typical Embedded System: Microprocessor vs. Microcontroller, RISC vs. CISC Processors, Harvard vs. Von-Neumann Processor Architecture, Big-Endian vs. Little-Endian Processors, Memory-ROM and RAM types, Sensors & Actuators, The I/O Subsystem – I/O Devices, Light Emitting Diode (LED), 7-Segment LED Display, Optocoupler, Relay, Piezo Buzzer, Push button switch.

Textbook 1: Ch. 1.1-1.6, Ch. 2.1-2.3

Module-2

Embedded System Design Concepts: Characteristics and Quality Attributes of Embedded Systems, Operational and Non-Operational Quality Attributes. Embedded Systems-Application and Domain Specific, Hardware Software Co-Design and Program Modelling.

Embedded Firmware Design and Development: Embedded Firmware Design Approaches, Embedded Firmware Development Languages.

Textbook 1: Ch. 3.1-3.2, Ch. 4.1-4.2 (4.2.1 and 4.2.2 only), Ch. 7.1-7.2, Ch. 9.1-9.2

Module-3

RTOS and IDE for Embedded System Design: Operating System Basics, Types of Operating Systems, Tasks, Process and Threads (Only POSIX Threads with an example program), Thread Preemption, Preemptive Task Scheduling Techniques, Task Communication, Task Synchronization Issues – Racing and Deadlock. How to Choose an RTOS, Integration and Testing of Embedded Hardware and Firmware.

Textbook 1: Ch. 10.1-10.3, 10.5.2, 10.7, 10.8.1.1, 10.8.1.2, 10.10, Ch. 12.1-12.2

Module-4

ARM Embedded Systems: The RISC Design Philosophy, The ARM Design Philosophy, Embedded System Hardware, Embedded System Software.

ARM Processor Fundamentals: Registers, Current Program Status Register, Pipeline, Exceptions, Interrupts, and the Vector Table, Core Extensions.

Textbook 2: Ch. 1.1-1.4, Ch. 2.1-2.5

Module-5

Introduction to the ARM Instruction Set: Data Processing Instructions, Branch Instructions, Load-Store Instructions, Software Interrupt Instruction, Program Status Register Instructions, Loading Constants.

Textbook 2: Ch. 3.1-3.6

Course Outcomes (Course Skill Set)

At the end of the course, the student will be able to:

1. Explain the characteristics and attributes of an Embedded System.
2. Illustrate the hardware software co-design and firmware design approaches of Embedded Systems.
3. Demonstrate the need of real time operating system for Embedded System applications.
4. Explain the ARM Architectural features and Instructions.
5. Develop programs using ARM instruction set for an ARM Microcontroller.

Assessment Details (both CIE and SEE)

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two assignments at the end of the semester if two assignments are planned.

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Semester-End Examination:

Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (**duration 03 hours).**

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3. The students have to answer 5 full questions, selecting one full question from each module.
4. Marks scored shall be proportionally reduced to 50 marks

Suggested Learning Resources:

Text Book:

1. Shibu K V, "Introduction to Embedded Systems", Second Edition, Tata McGraw Hill Education.
2. Andrew N Sloss, Dominic Symes and Chris Wright, "ARM System Developers Guide – Designing and Optimizing System Software", Elsevier, Morgann Kaufman Publishers.

Reference Books:

1. Raj Kamal, "Embedded Systems: Architecture and Programming", Tata McGraw Hill, 2008.
2. Raghunandan.G.H, "Microcontroller (ARM) and Embedded System", Cengage learning Publication, 2019.
3. "Insider's Guide to the ARM7 based microcontrollers", Hitex Ltd.,1st edition, 2005.

Web links and Video Lectures (e-Resources):

- https://alison.com/tag/embedded-systems
- https://www.youtube.com/watch?v=uFhDGagZzjs
- https://www.youtube.com/watch?v=G1c_WMD_5pU
- https://archive.nptel.ac.in/courses/106/105/106105193/

Activity Based Learning (Suggested Activities in Class)/ Practical Based learning

- Group task to demonstrate the Installation and working of Keil Software (10 Marks).
- Using Keil software, observe the various Registers, Dump, CPSR etc. and write Assembly Language Programs (15 Marks).